(a) Overview

Support-set adaptation

Labeled source support $\mathcal{S}$
2K images
K real …
K fake …

Trainable P

Four prompt groups
2 tokens per stage

*Save P**

CLIP ViT-L/14 blocks 1–24 ❄

image feature

Fixed t_0, t_1

Cosine logits
Native scale S

CE ← *y*

Once per support set

Gradients through frozen backbone; update P only

CLIP ViT-L/14 blocks 1–12 ❄

CLS
L_2 norm

Support cache
Support features $\{z_{ci}\}$ | Class prototypes μ_c

8,192 parameters optimized on support

Query inference · no parameter updates

Query x

Real + fakes from unseen generators

CLIP ViT-L/14 blocks 1–12 ❄

CLS
L_2 norm

$q \in \mathbb{R}^{1024}$

(b) Query-conditioned prototype mixing

$p_P(x)$

Fixed P*

*Load P**

(c) CLIP ViT-L/14 blocks 1–24 ❄

$v(x) \in \mathbb{R}^{768}$

Cosine difference
$v^T t_1 - v^T t_0$

Fixed t_0, t_1

$\times y \rightarrow \sigma$

$p_V(x)$

Probability average
$p(x) = \frac{1}{2}(p_P + p_V)$

p(x) > 0.5: Fake

(b) Prototype mixing

$q = f_{12}(x)$, $\|q\|_2 = 1$

Within-class weighting | **Weighted sum** | **Mix → L_2 norm**

Real
Fake
q

$\alpha_{ci}(q)$ → $m_0(q)$ → $\tilde{\mu}_0(q)$
$\alpha_{ci}(q)$ → $m_1(q)$ → $\tilde{\mu}_1(q)$

μ_0, μ_1 · 0.5 · 0.5 · L_2 norm

$$\alpha_{ci}(q) = \mathrm{softmax}_i(\beta\, q^T z_{ci})$$

$m_c(q)$ is not normalized before mixing.

$$p_P(x) = \sigma(\gamma\, q^T(\tilde{\mu}_1(q) - \tilde{\mu}_0(q)))$$

(c) Stage-wise prompt replacement

Frozen CLIP backbone ❄

P_1^* P_2^* P_3^* P_4^*

1–6 → 7–12 → 13–18 → 19–24 → CLIP CLS readout
768-D · normalized

CLS + image tokens continue

Output tokens → Drop prompt states → Insert P_{s+1}^*

CLS · state · image | Keep CLS + image | Next-stage input

Real · Fake · Prompts · ❄ Frozen